

According to the Markov property, a stochastic process X_t is said to satisfy the Markov property with respect to a filtration $\mathcal{F}_t$ if the conditional distribution of X_t given the past $\mathcal{F}_{t-1}$ depends only on the present state X_{t-1} . Formally, this can be expressed as:

$$\mathbb{P}(X_t \in A | \mathcal{F}_{t-1}) = \mathbb{P}(X_t \in A | X_{t-1})$$

for all measurable sets A . This means that the future state X_t is conditionally independent of the past states given the present state X_{t-1} .

0.0.1 Markov chains

A Markov chain is a sequence of random variables $(X_t)_{t \in \mathbb{N}}$ that satisfies the Markov property. The transition probabilities of a Markov chain are given by:

$$\mathbb{P}(X_{t+1} = x | X_t = y) = p(y, x)$$

where $p(y, x)$ is the transition probability from state y to state x . The Markov property implies that the future state X_{t+1} depends only on the present state X_t and not on the past states.

0.0.2 See also

0.0.3 References